Q4. [10 pts] PCA and least squares

Recall that PCA transforms (zero-mean) data into low-dimensional reconstructions that lie in the span of the top k eigenvectors of the sample covariance matrix. Let $\mathbf{U}_k$ denote the $d \times k$ matrix of the top k eigenvectors of the covariance matrix ($\mathbf{U}_k$ is a truncated version of $\mathbf{U}$, which is the matrix of eigenvectors of the covariance matrix).

There are two approaches to computing the low-dimensional reconstruction $\mathbf{w} \in \mathbb{R}^k$ of a data point $\mathbf{x} \in \mathbb{R}^d$:

1. Solve a least squares problem to minimize the reconstruction error
2. Project $\mathbf{x}$ onto the span of the columns of $\mathbf{U}_k$

In this problem, you will show that these approaches are equivalent.

(a) [5 pts] Formulate the least squares problem in terms of $\mathbf{U}_k, \mathbf{x}$, and the variable $\mathbf{w}$.

(Hint: This optimization problem should resemble linear regression.)

We want to find the weights such that when we weight the columns of U_k, we will minimize the residual error. Thus, the objective function is $\|U_k w - x_i\|^2$. Here is our least squares problem:

$$\min_w \|U_k w - x_i\|^2$$

(b) [5 pts] Show that the solution of the least squares problem is equal to $\mathbf{U}_k^\top \mathbf{x}$, which is the projection of $\mathbf{x}$ onto the span of the columns of $\mathbf{U}_k$.

Recall the normal equations (which is the most important equation in the class!). For some unconstrained least squares problem in the form

$$\min_x \|Ax - y\|^2$$

the solution of the minimizer is

$$x^* = (A^T A)^{-1} A^T y$$

only when A is full rank. This applies here since U_k is full rank by definition. Plugging in our least squares problem, we have

$$w^* = (U_k^T U_k)^{-1} U_k^T x_i$$

If you don't remember this you can easily derive this by taking the gradient of the objective function and setting it to 0. We note that $U_k^T U_k = I_k$, (I_k = the identity matrix in k dimensions) thus

$$w^* = (U_k^T U_k)^{-1} U_k^T x_i = (I_k)^{-1} U_k^T x_i = U_k^T x_i$$

since the inverse of I is simply I.

Q5. [10 pts] Mixture of linear regressions

In class, you learned that K-means partitions points into K clusters by minimizing an objective that encourages points to be close to their cluster centers. K-means minimizes this objective in a coordinate descent fashion, alternating between computing cluster assignments and cluster centers until convergence.

In this problem, you will devise an algorithm, in the spirit of K-means, that fits an entirely different kind of mixture model. You are given a dataset $\{\mathbf{x}_i, y_i\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. You know that this dataset is a mixture of realizations of K different linear regression models

$$y = \mathbf{w}_k^\top \mathbf{x} + \mathcal{N}(0, \sigma^2)$$

parameterized by K weight vectors $\mathbf{w}_1, \ldots, \mathbf{w}_K \in \mathbb{R}^d$.

Your algorithm will jointly determine the following:

- A partition $S_1, \ldots, S_K$ of the dataset such that $(\mathbf{x}_i, y_i) \in S_k$ iff $(\mathbf{x}_i, y_i)$ comes from model k
- The model weights $\mathbf{w}_1, \ldots, \mathbf{w}_K$

(a) [4 pts] Write an objective function $f(S_1, \ldots, S_K, \mathbf{w}_1, \ldots, \mathbf{w}_K)$ to be minimized to solve this problem. Use the penalty $\|\mathbf{w}_k^\top \mathbf{x} - y\|^2$ if the point $(\mathbf{x}, y)$ is assigned to model k. Your objective should be a sum of N terms, and each data point should show up in exactly one of these terms.

$$f(S_1, \ldots, S_K, w_1, \ldots, w_K) = \sum_{k=1}^{K} \sum_{(x,y) \in S_k} \|w_k^\top x - y\|^2$$

(b) [3 pts] What is coordinate descent update for f with $\mathbf{w}_1, \ldots, \mathbf{w}_K$ fixed? In other words, to which of the K models should a point $(\mathbf{x}, y)$ be assigned?

Assign the point (x, y) to S_k if $k = \arg\min_k \|w_k^\top x - y\|^2$.

(c) [3 pts] Write the coordinate descent update for f with $S_1, \ldots, S_K$ fixed.

For a set S of $(\mathbf{x}, y)$ values, you should use the notation $\mathbf{X}_S$ to denote the design matrix whose rows are the $\mathbf{x}$-values of the elements of S, and $\mathbf{y}_S$ to denote the column vector of y-values of the elements of S.

$w_k = (X_{S_k}^\top X_{S_k})^{-1} X_{S_k}^\top y_{S_k}$

Q6. [10 pts] Training set augmentation

In class, you learned that one way to encourage invariance of a model to certain transformations is to augment the training set with extra examples perturbed according to those transformations. In this problem, you will examine the behavior of a certain type of input perturbation for a probabilistic linear regression setting.

Consider the following general generative model for regression:

- $\mathbf{x} \sim p(\mathbf{x})$, where $p(\mathbf{x})$ is a distribution over input vectors $\mathbf{x} \in \mathbb{R}^d$
- $y|\mathbf{x} \sim p(y|\mathbf{x})$, where $p(y|\mathbf{x})$ is a distribution over output scalars $y \in \mathbb{R}$ given $\mathbf{x}$

Assume that the relationship between y and $\mathbf{x}$ is well-modeled by a linear function $y = \mathbf{w}^\top \mathbf{x}$, where $\mathbf{w} \in \mathbb{R}^d$, so that in the infinite dataset limit, the objective to be minimized for this regression problem is:

$$\mathcal{L}_0(\mathbf{w}) = \mathbb{E}\left[(\mathbf{w}^\top \mathbf{x} - y)^2\right]$$

Now suppose the inputs are perturbed by zero-mean Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$, which is independent of the training data. The new objective is

$$\mathcal{L}(\mathbf{w}) = \mathbb{E}\left[(\mathbf{w}^\top (\mathbf{x} + \boldsymbol{\epsilon}) - y)^2\right]$$

(a) [9 pts] Compute and simplify $\mathcal{L}(\mathbf{w})$. Show all your work in detail, and write your answer in terms of $\mathcal{L}_0$.

$$\begin{aligned}
\mathcal{L}(w) &= \mathbb{E}\left[(w^\top (x + \epsilon) - y)^2\right] \\
&= \mathbb{E}\left[((w^\top x - y) + w^\top \epsilon)^2\right] \\
&= \mathbb{E}\left[(w^\top x - y)^2 + 2(w^\top x - y) w^\top \epsilon + w^\top \epsilon \epsilon^\top w\right] \\
&= \mathbb{E}\left[(w^\top x - y)^2\right] + \mathbb{E}\left[2(w^\top x - y) w^\top\right] \mathbb{E}[\epsilon] + w^\top \mathbb{E}\left[\epsilon \epsilon^\top\right] w
\end{aligned}$$

Substituting $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon \epsilon^\top] = \lambda I$, this simplifies to

$$= \mathcal{L}_0(w) + \lambda \|w\|^2$$

(b) [1 pt] Is there a relationship between this particular type of input perturbation and some type of regularization? If so, what kind of regularizer is involved?

In this setting, regression assuming this type of input perturbation turns out to be equivalent to regression with an L_2 regularizer.

Q7. [12 pts] Kernel PCA

You are given d-dimensional real-valued data $\{\mathbf{x}_i\}_{i=1}^N$ and a feature mapping $\phi : \mathbb{R}^d \to \mathbb{R}^m$. In the following questions, you will investigate how to do PCA in feature space on the feature vectors $\{\phi(\mathbf{x}_i)\}_{i=1}^N$. Assume that the data is centered in feature space; that is, $\sum_{i=1}^N \phi(\mathbf{x}_i) = 0$.

In the following, $\mathbf{\Phi}$ is a design matrix whose i^{th} row is $\phi(\mathbf{x}_i)$.

(a) [1 pt] Recall that as a part of PCA, we must solve the eigenvalue problem $\mathbf{Sv} = \lambda\mathbf{v}$, where $\mathbf{S}$ proportional to the sample covariance matrix. For PCA in feature space, we have $\mathbf{S} = \sum_{i=1}^N \phi(\mathbf{x}_i)\phi(\mathbf{x}_i)^\top = \mathbf{\Phi}^\top\mathbf{\Phi}$. Why is this a problem if m is large?

Working in feature space directly is too expensive if m is large. The covariance matrix $\Phi^\top\Phi$ is $m \times m$, which is too large to compute and work with.

(b) [3 pts] Now, you are given a kernel function $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}')$. Define the kernel matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.

Show that if $\lambda \neq 0$, then λ is an eigenvalue of $\mathbf{S}$ if and only if λ is also an eigenvalue of $\mathbf{K}$ (in other words, finding feature-space principal components can be done by finding eigenvectors of $\mathbf{K}$).

Notice that $K = \Phi\Phi^\top$. The nonzero eigenvalues of $K = \Phi\Phi^\top$ and $S = \Phi^\top\Phi$ are the same.

(c) [4 pts] Let $\mathbf{v}$ be an eigenvector of $\mathbf{S}$ with nonzero eigenvalue λ. Show that $\mathbf{v}$ can be written as $\mathbf{v} = \mathbf{\Phi}^\top\boldsymbol{\alpha}_\mathbf{v}$, where $\boldsymbol{\alpha}_\mathbf{v}$ is an eigenvector of $\mathbf{K}$ with eigenvalue λ.

First, write $Sv = \Phi^\top\Phi v = \lambda v$. Multiplying this equation by Φ gives $K\alpha = \lambda\alpha$, where $\alpha = \Phi v$. Then, since $\Phi^\top\alpha = \Phi^\top\Phi v = Sv = \lambda v$, we have $v = \Phi^\top\alpha/\lambda$. Choosing $\alpha_v = \alpha/\lambda$ gives the desired result.

(d) [4 pts] You are given a new data point $\mathbf{x} \in \mathbb{R}^d$. Find the scalar projection of its feature representation $\phi(\mathbf{x})$ onto $\mathbf{v}/\|\mathbf{v}\|$ (with $\mathbf{v}$ defined as above).

Write your answer in terms of $\boldsymbol{\alpha}_\mathbf{v}$ and λ. Use the kernel k, and do not explicitly use ϕ. You should use the notation $\mathbf{k}_\mathbf{x} = \begin{bmatrix} k(\mathbf{x}_1, \mathbf{x}) & \cdots & k(\mathbf{x}_n, \mathbf{x}) \end{bmatrix}^\top$.

First, let's calculate the squared norm of v:

$$\|v\|^2 = v^\top v = \alpha_v^\top \Phi\Phi^\top \alpha_v = \alpha_v^\top K \alpha_v = \lambda\|\alpha_v\|^2$$

Then, the projection is:

$$\frac{v^\top\phi(x)}{\|v\|} = \frac{\alpha_v^\top\Phi\phi(x)}{\sqrt{\lambda}\|\alpha_v\|} = \frac{\alpha_v^\top k_x}{\sqrt{\lambda}\|\alpha_v\|}$$